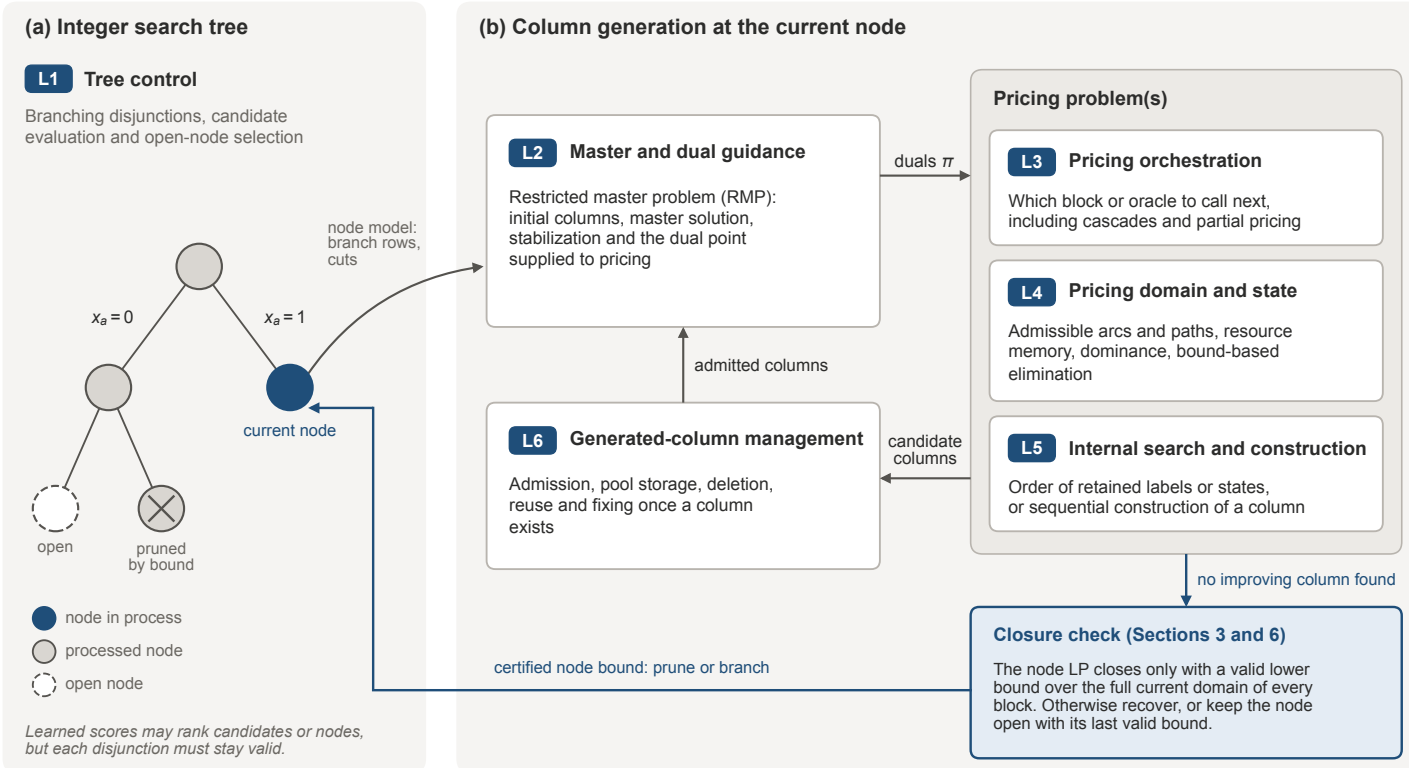

(a) Integer search tree
L1 Tree control
Branching disjunctions, candidate evaluation and open-node selection
node model: branch rows, cuts
$x_a = 0$
$x_a = 1$
current node
open
pruned by bound
node in process
processed node
open node
Learned scores may rank candidates or nodes, but each disjunction must stay valid.
(b) Column generation at the current node
L2 Master and dual guidance
Restricted master problem (RMP): initial columns, master solution, stabilization and the dual point supplied to pricing
duals π
Pricing problem(s)
L3 Pricing orchestration
Which block or oracle to call next, including cascades and partial pricing
L4 Pricing domain and state
Admissible arcs and paths, resource memory, dominance, bound-based elimination
L5 Internal search and construction
Order of retained labels or states, or sequential construction of a column
candidate columns
admitted columns
L6 Generated-column management
Admission, pool storage, deletion, reuse and fixing once a column exists
no improving column found
Closure check (Sections 3 and 6)
The node LP closes only with a valid lower bound over the full current domain of every block. Otherwise recover, or keep the node open with its last valid bound.
certified node bound: prune or branch